AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 14 · STUDENT LESSON

Build a Human Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Build the whole classifier, including rejection.

Give Tower1 an input contract, inspect worked RibbonR, operate a fictional creature classifier and construct a separate Ribbon1.

Keep one habit: Validate the record → predict only when accepted → retain the version and outcome.

MAIN LESSON · SECTION 1

Can someone operate your classifier without guessing?

> Validate the record before applying the selected rule, and keep rejection separate from a category.

Start with the whole procedure

You will act as the operator of a paper classifier for fictional measurements. A threshold is only one part of the procedure: you also need accepted inputs, a rejected-input path and a record of which model was intended.

Retrieve the missing-height question

A fitted rule predicts tall at a height of at least 6 cm and short otherwise. Predict at 5 and 6 cm. What should you check before applying it to a blank height?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can construct a **classifier** with a stated **input** format, selected rule and recorded **output**.
- I can trace valid, boundary and invalid **input**s through different paths.
- I can distinguish an **input** check from **evidence** that a **prediction** is correct.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Classification maps features to a category using a **decision boundary**.
</details>



LEARNING TARGET A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can build a complete **classifier** that validates **input**s before predicting.

MAIN LESSON · SECTION 2

Give Tower1 a complete input contract

> The model uses the threshold selected in Week 13's updated tower fit.

Identify Tower1 and where its rule came from

Your task is to build a paper **classifier** that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers.

Use the threshold of 6 centimetres selected after adding the height 5 short example in Week 13. Name this selected model Tower1.

State accepted records and endpoints

Write an **input contract**: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts.

It is not a claim that every accepted height appeared in **fitting** or that towers outside this range cannot exist.

Include the rejected-input path

A complete **classifier** needs instructions for checking the **input**, applying Tower1 when the **input** is accepted, and recording the outcome. It must also say what happens when the **input** fails a check.

The threshold alone does not provide these surrounding instructions.

Check the instructions another learner needs

- Name the selected model, threshold, required unit and accepted range. Which instructions must still surround the threshold before another learner can operate the procedure?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY Mystery creatures can be sorted using visible features such as antenna count and spot pattern, with the decision rule drawn on a feature map.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Validate, stop if rejected, then predict

> Record the model version even when its threshold was not reached.

Trace accepted and missing-height paths

Three records through the Tower1 classifier

INPUT RECORD	CHECK AND RULE	RECORDED OUTCOME
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

Apply the input checks in order

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10.

If any check fails, record **no prediction** and the reason, then stop **process**ing that record. Do not continue into the threshold comparison.

Predict only after every check passes

If all checks pass, compare the height with 6. At least 6 gives the **prediction** tall; otherwise give short. Record the original **input**, **model version** Tower1 and the **prediction**.

For a rejected record, record the original **input**, Tower1, **no prediction** and a reason. Tower1 identifies the intended model even when its threshold was not used.

Explain the stopping branch

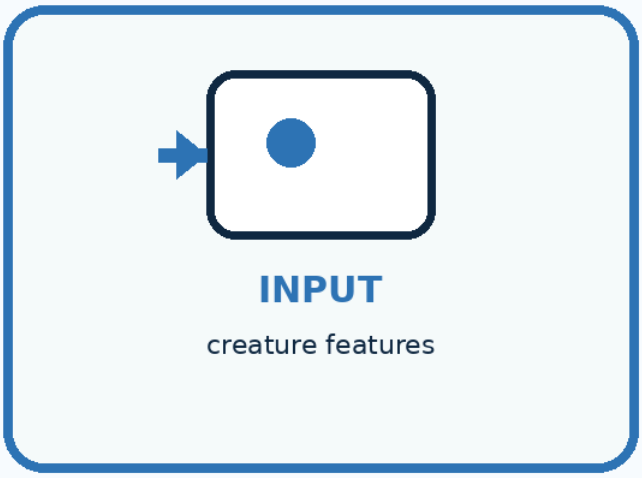
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

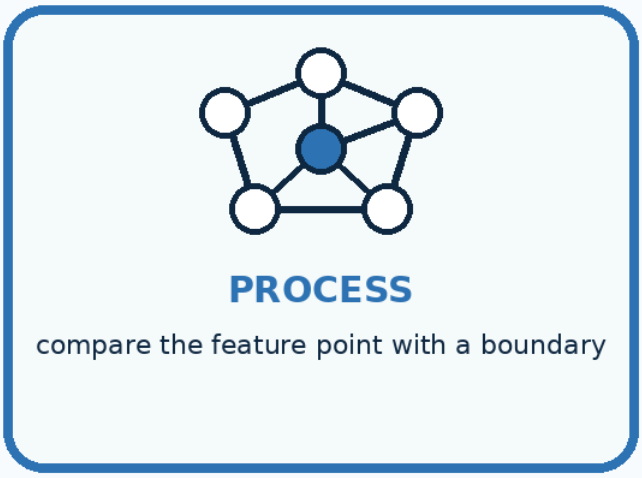
▼ Earlier wording, alternatives and picture transcripts

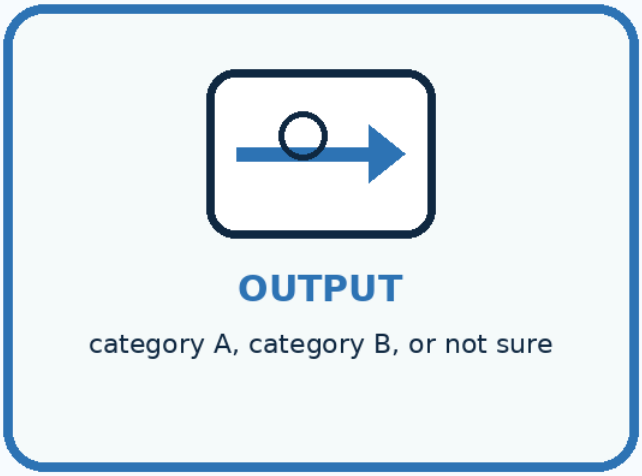
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT creature features

PROCESS compare the feature point with a boundary

OUTPUT category A, category B, or not sure

CORE IDEA Classification maps features to a category using a **decision boundary**.

UNDER THE HOOD A **classifier** partitions **feature space** into decision regions. A **decision boundary** is where the predicted category changes.

Different boundaries may fit the same **training** examples but behave differently on unseen or borderline **input**s.

Draw a fresh example with all three stages:

A closer explanation

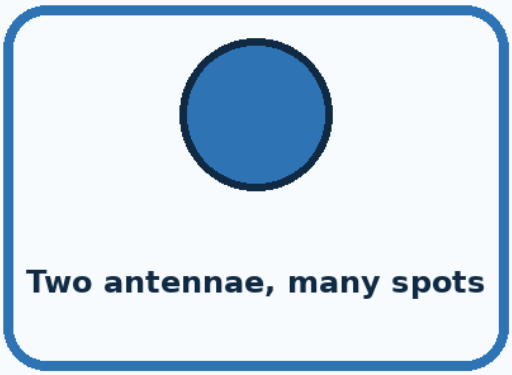
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare classifier inputs and decision clues

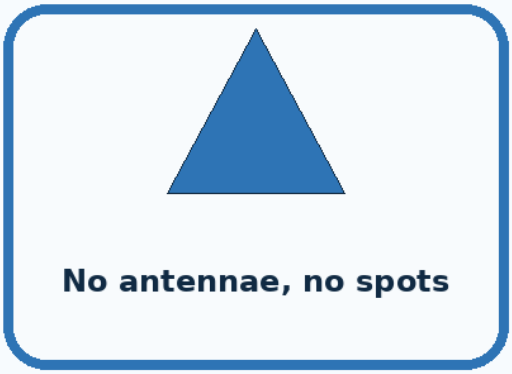
> Keep a feature, the declared boundary and missing evidence distinguishable.

Pictures paired with their evidence



The case to investigate

CASE 1 — Two antennae, many spots Clue: far inside region A. Meaning: stable A **prediction**.



The case to investigate

CASE 2 — No antennae, no spots Clue: far inside region B. Meaning: stable B **prediction**.



The case to investigate

CASE 3 — One antenna, some spots Clue: near boundary. Meaning: uncertain.



The case to investigate

CASE 4 — New wing feature Clue: not represented on map. Meaning: boundary may be incomplete.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

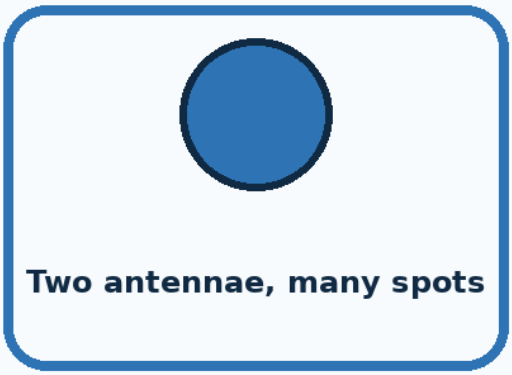
BOUNDARY A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

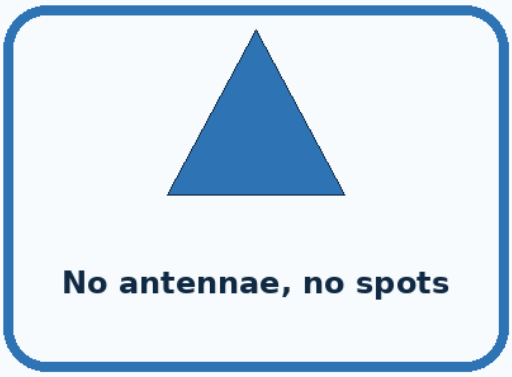
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Two antennae, many spots Clue: far inside region A. Meaning: stable A **prediction**.

CASE 2 — No antennae, no spots Clue: far inside region B. Meaning: stable B **prediction**.

CASE 3 — One antenna, some spots Clue: near boundary. Meaning: uncertain.

CASE 4 — New wing feature Clue: not represented on map. Meaning: boundary may be incomplete.

Change one thing in the pictured case

Change: Provide length 9 with unit mm.

Prediction: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

MAIN LESSON · SECTION 6

RibbonR checks its contract before its boundary

> This fixed instruction card accepts whole lengths 0–15 cm and predicts long at 9 cm or more.

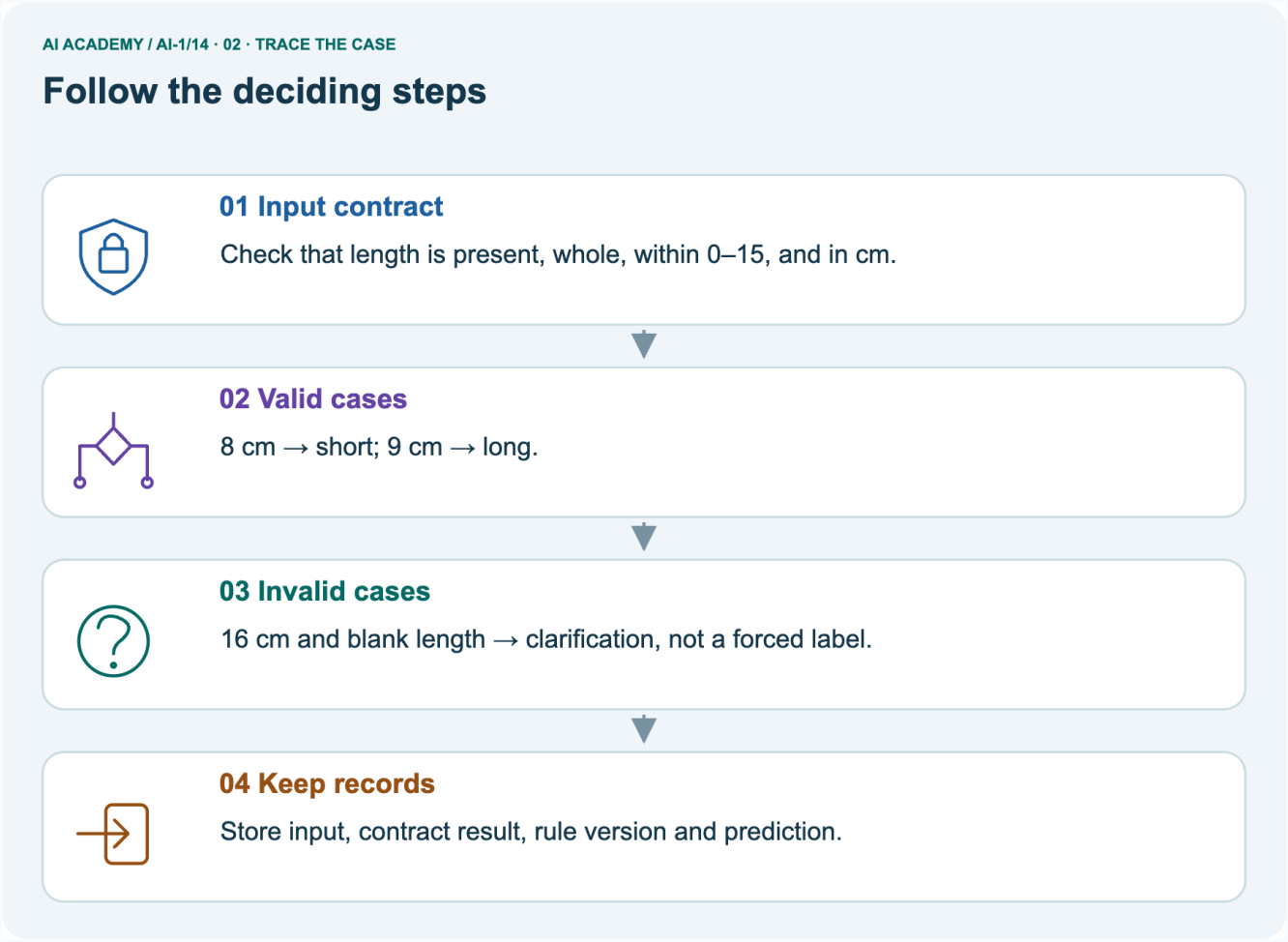
The complete RibbonR card

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Valid **input**: whole-number length from 0 to 15 with unit cm.

Predict long at least 9 cm; otherwise short. Missing, wrong-unit or out-of-range values request clarification.

Trace validation, prediction and the saved record



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the stated outcomes

AI ACADEMY / AI-1/14 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why 16 is rejected while 0 is accepted

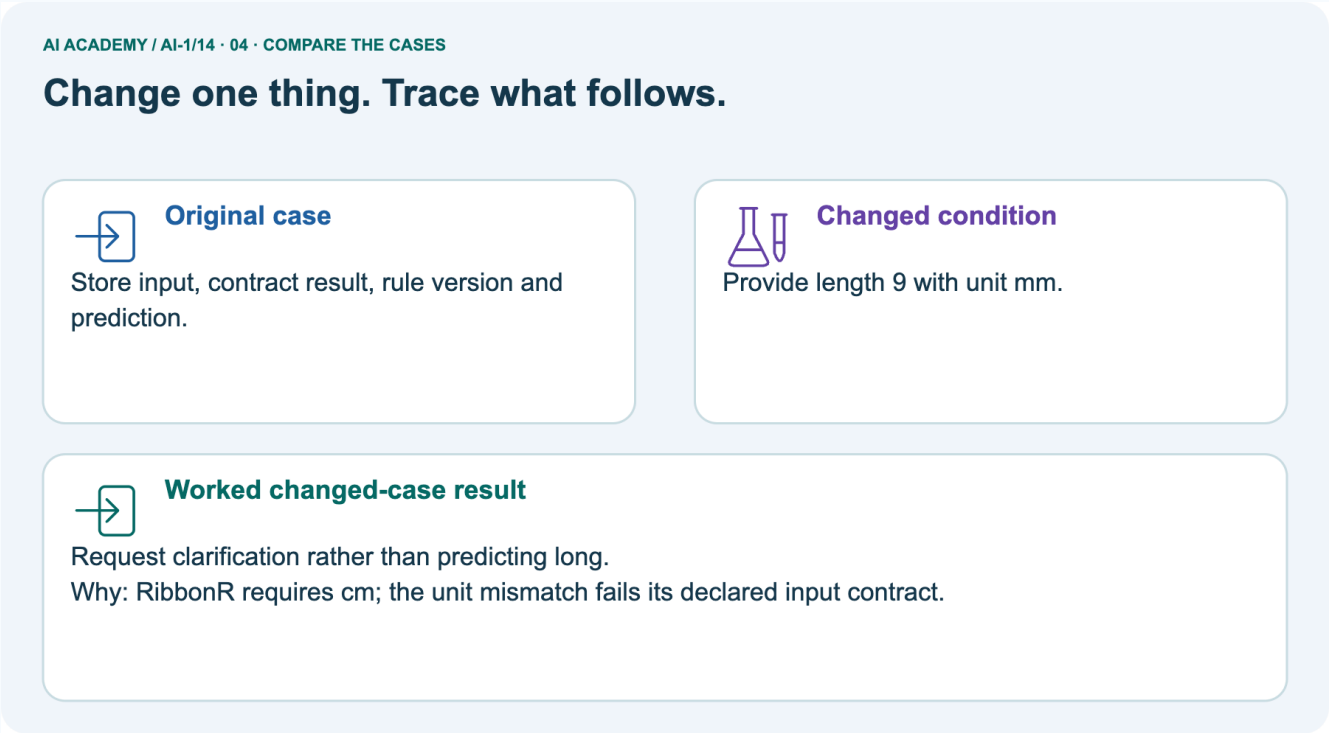
A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Predict with the wrong unit

- Before checking, supply the digits 9 with unit mm. Use RibbonR’s cm-only contract. Does the threshold comparison run? Explain the next action without silently changing units.

▼ Check the unit-mismatch path



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep validation separate from correctness

Contract validity does not prove the predicted category agrees with a checked reference label.

Explain 0 cm and clarification

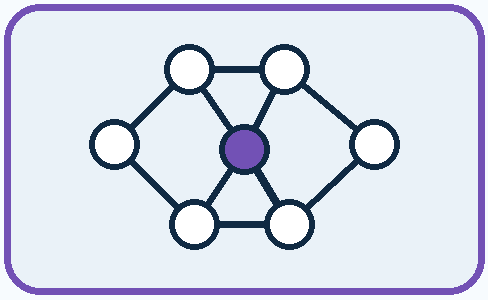
- Read the labelled picture: Trace 0 cm and explain why clarification is different from predicting short.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE Mystery creatures can be sorted using visible features such as antenna count and spot pattern, with the decision rule drawn on a feature map.

STEP 1 — OBSERVE creature features

STEP 2 — TRACE compare the feature point with a boundary

STEP 3 — CONCLUDE category A, category B, or not sure

MODEL REASONING Plot each creature by the selected features and draw a line or rule separating labeled examples. Cases near the line are sensitive to small changes and should receive careful review or a not-sure **output**.

DO NOT OVERCLAIM A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

What new **evidence** could change this answer?

A closer explanation

Input	Contract result	Outcome
1 cm	Accepted at lower endpoint	Predict short
9 cm	Accepted	Predict tall
10 cm	Accepted at upper endpoint	Predict tall
0 cm	Outside 1 through 10	No prediction

Input	Contract result	Outcome
60 mm	Unit is not cm	No prediction
4.5 cm	Not a whole number	No prediction

A valid 9 cm **input** receives a tall **prediction**, but the **input** check does not establish that the **prediction** matches its comparison label. **Validation** checks the contract.

Evaluation checks **prediction**s against appropriate **evidence**. Keep these questions separate.

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step.

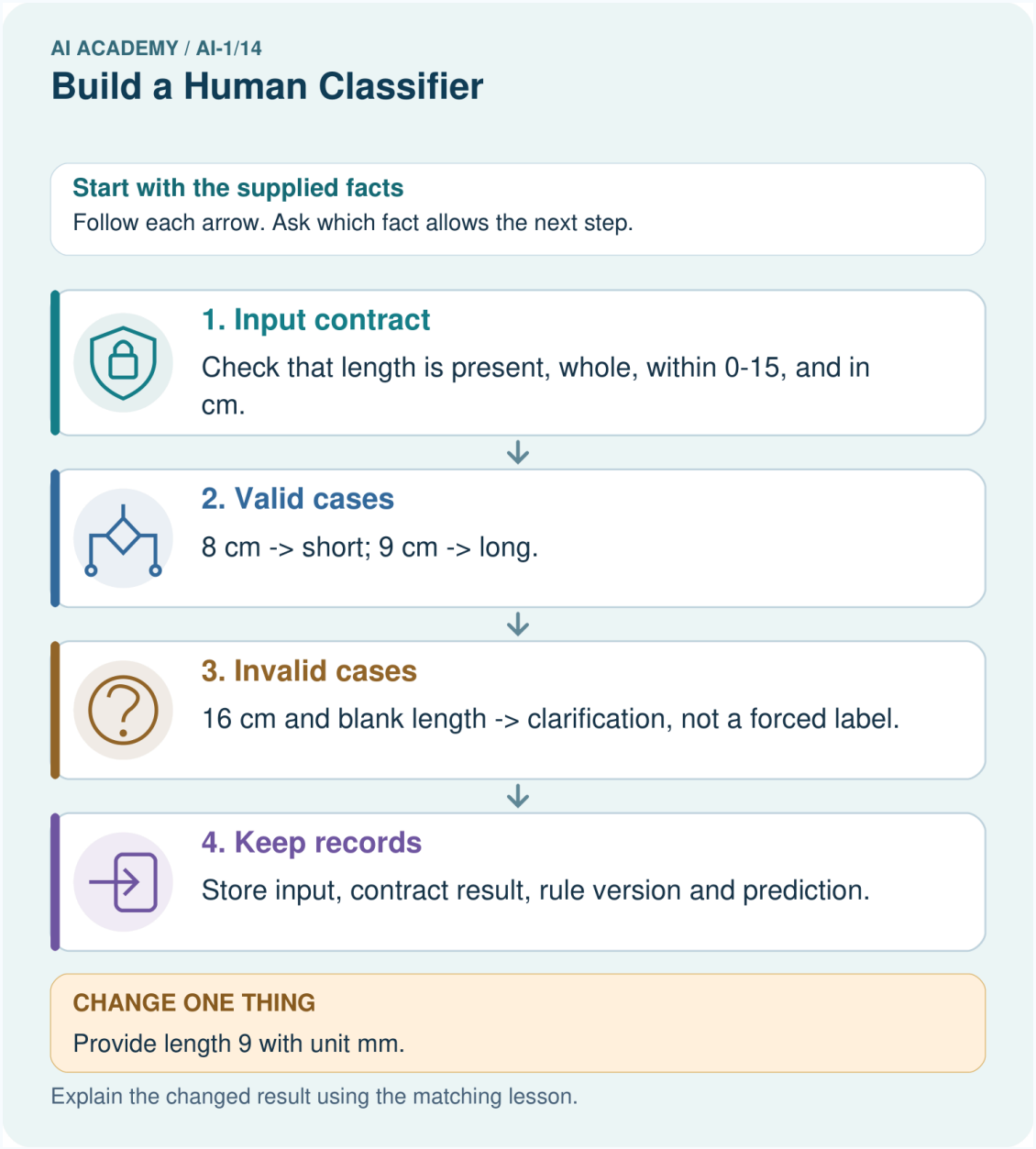
Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Build a Human Classifier

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Input contract	Check that length is present, whole, within 0–15, and in cm.
2. Valid cases	8 cm → short; 9 cm → long.
3. Invalid cases	16 cm and blank length → clarification, not a forced label.
4. Keep records	Store input, contract result, rule version and prediction.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Provide length 9 with unit mm. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

No prediction is a recorded system outcome

> It is neither a hidden short prediction nor a decision about what a person must do.

Keep the later action separate

The **classifier** reports a **prediction** or **no prediction**. It does not decide what a person must do with a tower.

A separate classroom procedure might ask someone to remeasure a rejected **input** or compare a **prediction** with a recorded label. Write that action separately from the predicted category.

Retain rejected records in the system evidence

No prediction is not a third tower size. It records that this procedure did not assign short or tall.

Count and investigate such records when evaluating the whole system; hiding them could make a system look more useful than it is.

Repair the validation-equals-correctness claim

- CLAIM A Anything on the same side must truly belong to that category.
- CLAIM B Only one possible boundary can fit the examples.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A classifier accepts 9 cm and predicts tall. A learner says the input passed, so the label is definitely right. Correct the claim. Then explain why reporting short for a blank height would violate the Tower1 procedure**

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Two antennae, many spots	far inside region A	stable A prediction
No antennae, no spots	far inside region B	stable B prediction
One antenna, some spots	near boundary	uncertain
New wing feature	not represented on map	boundary may be incomplete

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: hand the procedure to another operator

> A partner should follow written branches and stop points without inventing missing instructions.

Construct and test Tower1

Write the procedure as numbered steps or connected boxes with labelled branches. Give it to a partner who follows only what you wrote.

The partner should record each check and stop point, including equality at the threshold and at both accepted range endpoints.

> **Construct Tower1 in your own words. Have a partner trace 1 cm, 6 cm, 10 cm, 11 cm and a blank height. Record each outcome and any instruction you had to clarify. Explain why the accepted 10 cm case still needs separate evidence of correctness**

Keep instruction repairs distinct from model changes

A partner finding an unclear branch gives useful design feedback. Revise the instruction and retrace the affected examples.

Keep the model name and threshold visible so a change to the fitted rule is not confused with a correction to an instruction.

Use W2 to write both outcome paths

Use workbook W2 for the connected example “Constructing a **classifier**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “every numeric input gets a category”

- Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to RibbonR’s zero case

Trace 0 cm and explain why clarification is different from predicting short.

> **My explanation using a pictured step and the given facts**

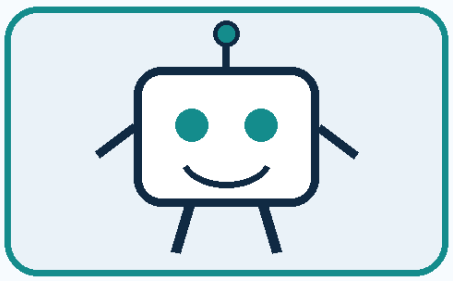
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Classification maps features to a category using a **decision boundary**.

Use a fresh example to explain **classifier**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Only one possible boundary can fit the examples.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Construct both outcome paths

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate a complete mystery-creature instruction card

> Use a fictional count contract, an equality case and explicit rejected-input records.

The complete CreatureC policy

Use fictional creature cards with an antenna-count field. CreatureC accepts one whole-number count from 0 through 4. If missing, unreadable, fractional or outside that range, record NO PREDICTION with a reason and stop. Otherwise predict A when count is at least 2 and B otherwise. Record the original count, contract result, CreatureC and the prediction or reason. A and B are invented task labels.

Predict, operate and compare records

1. Predict and trace cards showing counts 1, 2, 5 and blank. Some pictures have stripes and others dots; pattern is not an input to this declared rule. Have a partner follow only your instruction card. Keep the invalid 5 case out of the threshold comparison, and explain why a valid A prediction still needs separate comparison evidence before claiming correctness.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep contract and prediction evidence separate

Case	Prediction	Observed result	What it means
Normal			
Edge			

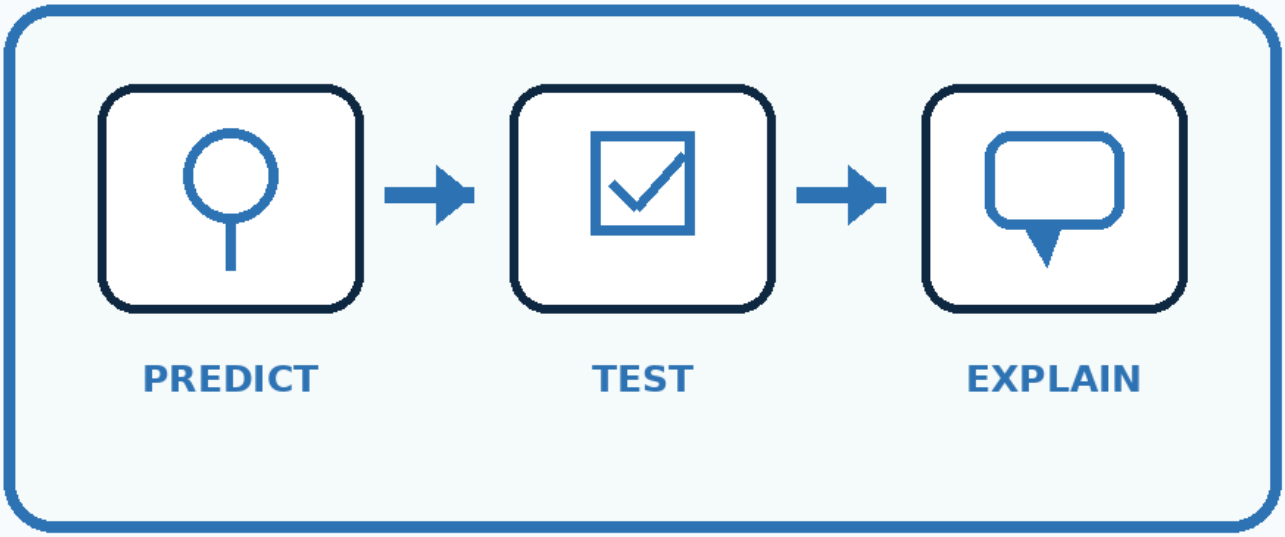
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Mystery Creature Classifier

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Classify mystery creatures and make the boundary visible.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: construct the separate Ribbon1 classifier

> Use Ribbon1's own range 1–12 cm and long boundary 8; keep its name with every outcome.

The new input contract and requested traces

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

My independent explanation, trace and limits

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build an independent ribbon classifier

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use a human **classifier**: three straight sides forming a closed shape means triangle; four equal straight sides and four right angles forming a closed shape means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Make the sorter's current **decision boundary** visible as a rule chart or feature map.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Write the complete sorter instruction card

> Include accepted formats, the prediction rule, rejection reasons and who resolves clarification requests.

This week’s project decision

THIS WEEK'S CHECKPOINT Make the sorter's current **decision boundary** visible as a rule chart or feature map.

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

Plan input checks, tests and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep the decision boundary’s meaning limited

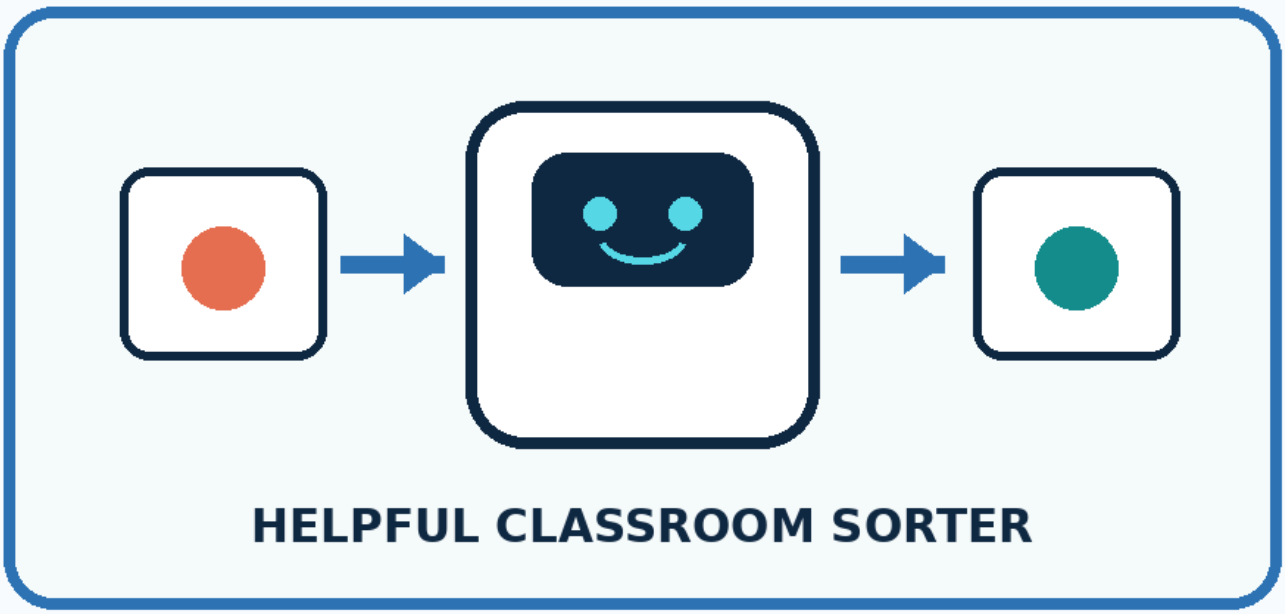
RESPONSIBLE DESIGN A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

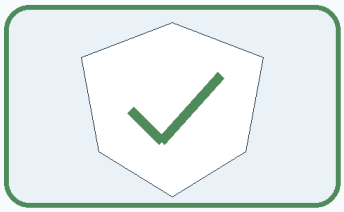
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: distinguish validation from evaluation

> Close your notes and explain an accepted input, a rejected input and the remaining correctness question.

Your exit explanation

- Close your notes. Explain this goal in your own words: Construct a **classifier** with a stated **input** format, selected rule and recorded **output**

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
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Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
classifier	process that predicts a category	
feature space	map whose positions represent feature values	
decision boundary	place where predicted class changes	
borderline case	input near or across competing regions	

Terms for the connected explanation

Word	Meaning
Input contract	The format and conditions an input must satisfy for a procedure to accept it
Validation	Checking whether an input satisfies the stated contract
No prediction	An explicit outcome when the classifier does not assign a category

Word	Meaning
Model version	A name that identifies the particular model used for a prediction

Repair a claim using a word

- CLAIM TO REPAIR Anything on the same side must truly belong to that category.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

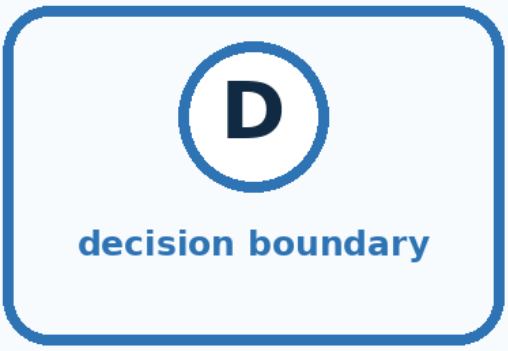
VOCABULARY

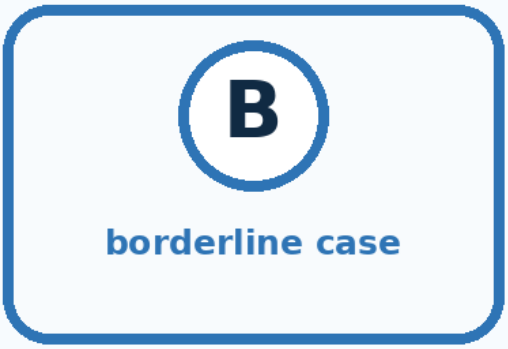
Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

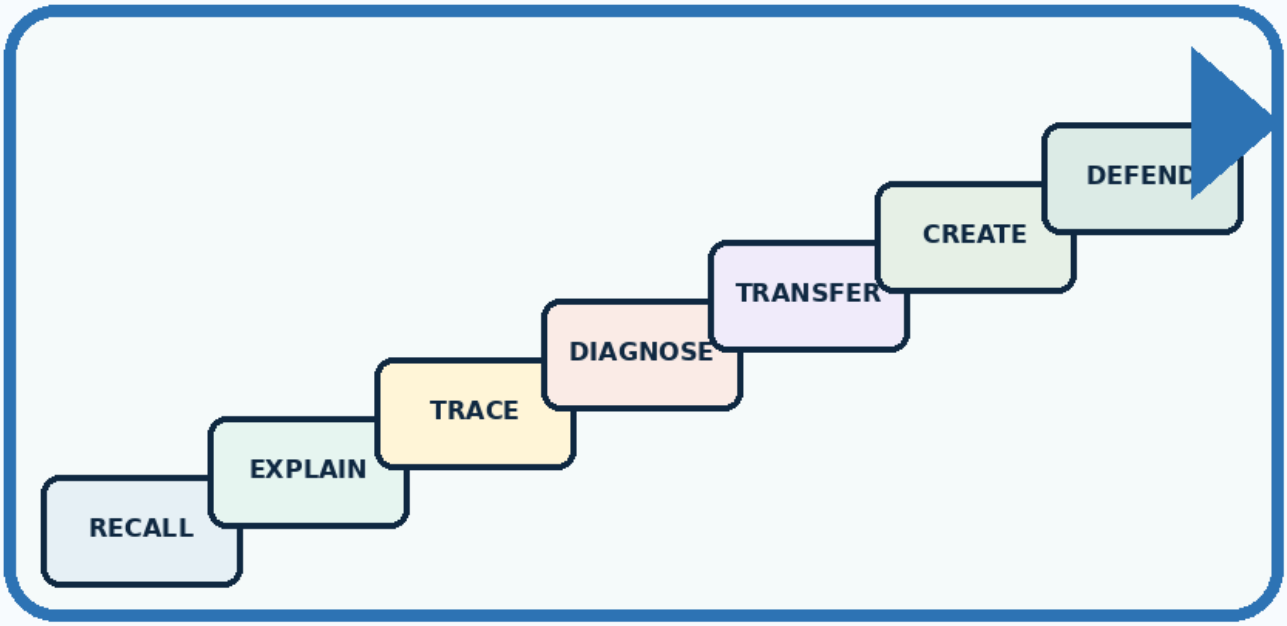
- RECALL Define **classifier**.
- EXPLAIN Classification maps features to a category using a **decision boundary**.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Anything on the same side must truly belong to that category.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Build an independent ribbon classifier is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

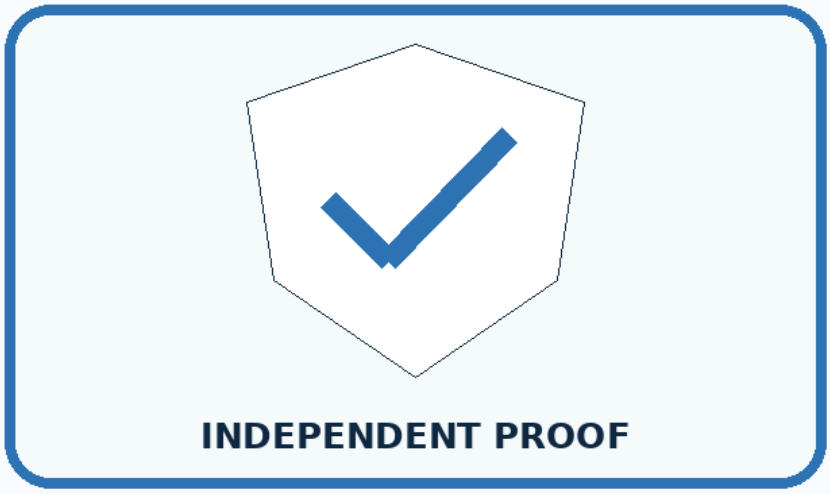
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Build an independent ribbon classifier is the selected independent case.



EXPLANATION CHECK Validate against the stated contract, stop rejected records, retain the model version and distinguish input acceptance from evidence of a correct prediction.

Independent **evidence**:

SELF-CHECK Plot each creature by the selected features and draw a line or rule separating labeled examples. Cases near the line are sensitive to small changes and should receive careful review or a not-sure **output**.

NEXT BRIDGE Carry today's **evidence** habit into: Save New Examples for Testing.

▼ **Extra practice: Does the object have wheels?**

REFERENCE · SECTION 13

Extra practice: Does the object have wheels?

> Try the question before opening its worked explanation.

Read the supplied case and question

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

Apply the rule and identify a limitation.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

Evidence or stage	Value or result
Bicycle	Yes → vehicle
Wheelchair	Yes → vehicle
Toy car	Yes → vehicle
Boat	No → excluded by this rule

▼ Earlier wording, alternatives and picture transcripts

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Does the object have wheels?

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

Apply the rule and identify a limitation.

The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Construct a classifier with a stated input format, selected rule and recorded output	Explain the deciding step, show a trace or test, and state one limit.
Trace valid, boundary and invalid inputs through different paths	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an input check from evidence that a prediction is correct	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional ribbon sorter uses a fixed instruction card called RibbonR.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional ribbon sorter uses a fixed instruction card called RibbonR.
- Which labelled step matches this detail? Store **input**, contract result, rule version and **prediction**.
- 1. Invalid cases
- 2. **Input contract**
- 3. Keep records
- 4. Valid cases
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Keep records Store **input**, contract result, rule version and **prediction**. A **prediction** boundary answers a different question from **input validation**.

Length 9 passes the contract and meets the long boundary. Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Changed-case question

- Changed-case practice: Provide length 9 with unit mm. Which conclusion is supported by the supplied facts?
- 1. Request clarification rather than predicting long.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Request clarification rather than predicting long. RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Repair and save your work

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

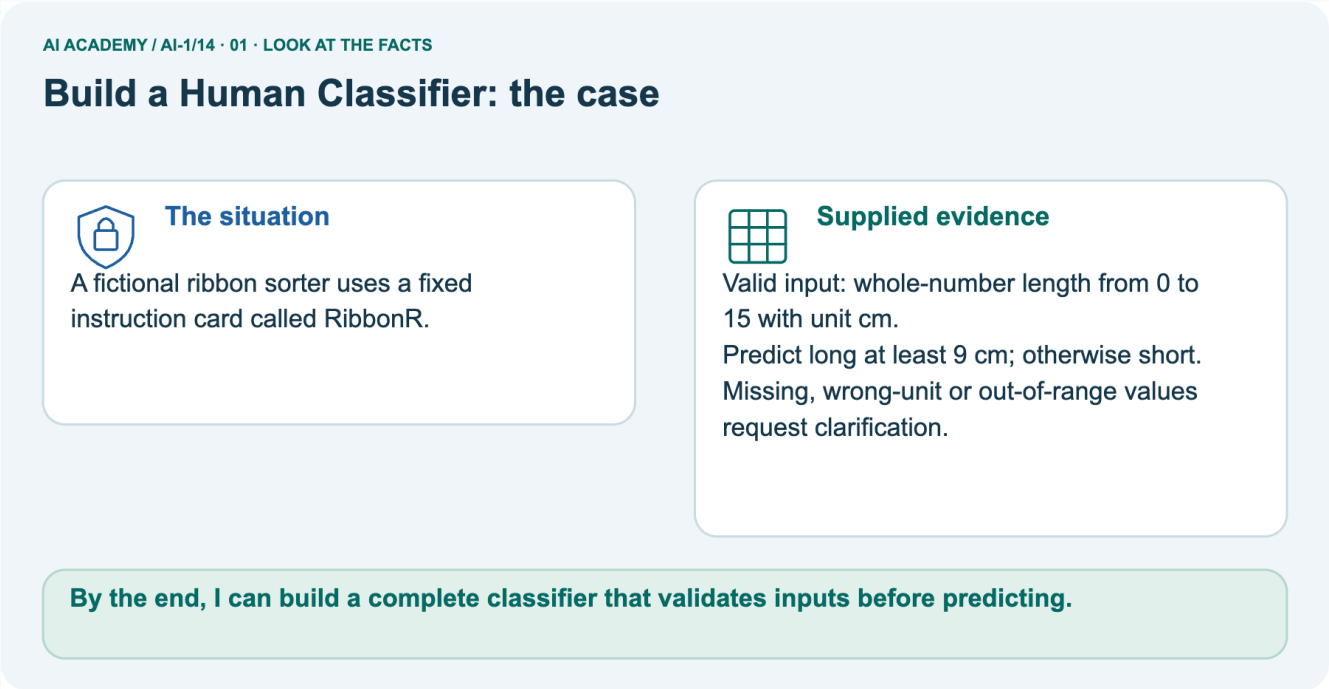
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Build a Human Classifier: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional ribbon sorter uses a fixed instruction card called RibbonR. Supplied facts: Valid **input**: whole-number length from 0 to 15 with unit cm. Predict long at least 9 cm; otherwise short.

Missing, wrong-unit or out-of-range values request clarification.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

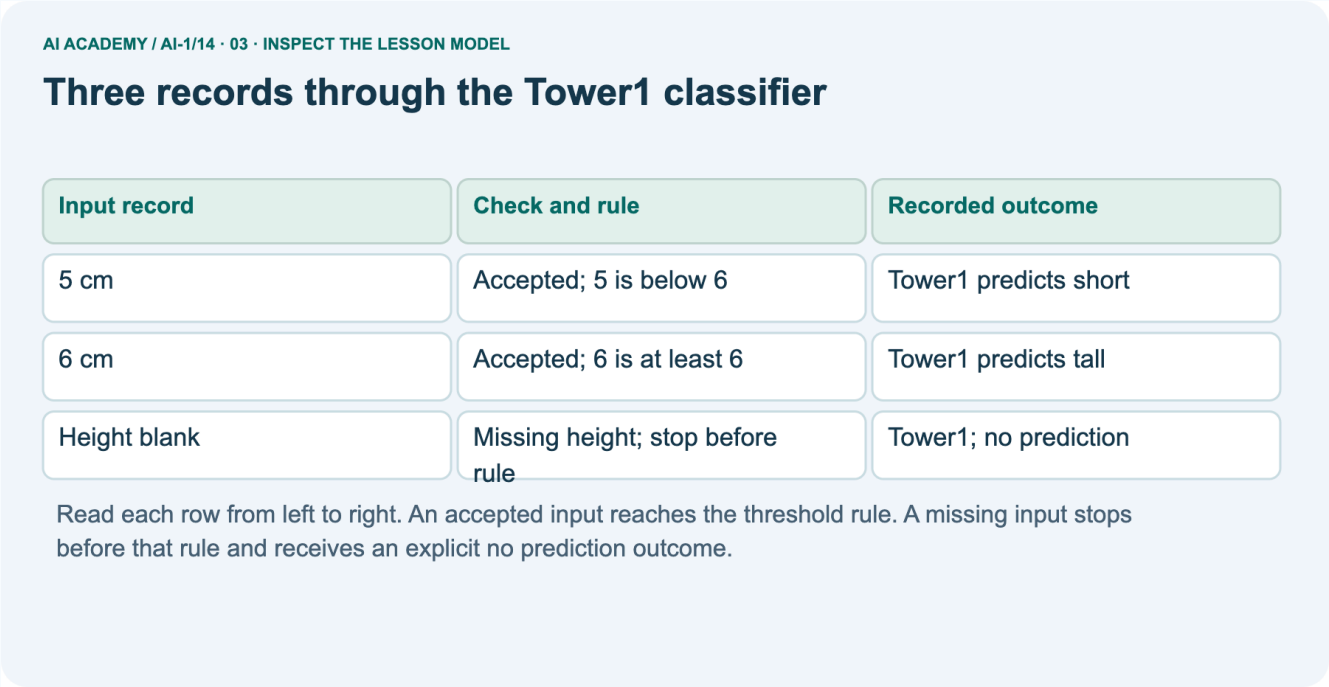


Follow the labelled picture with the matching explanation. Open the vector diagram

Input contract: Check that length is present, whole, within 0–15, and in cm. → Valid cases: 8 cm → short; 9 cm → long. → Invalid cases: 16 cm and blank length → clarification, not a forced label. → Keep records: Store **input**, contract result, rule version and **prediction**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Three records through the Tower1 classifier



AI ACADEMY / AI-1/14 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

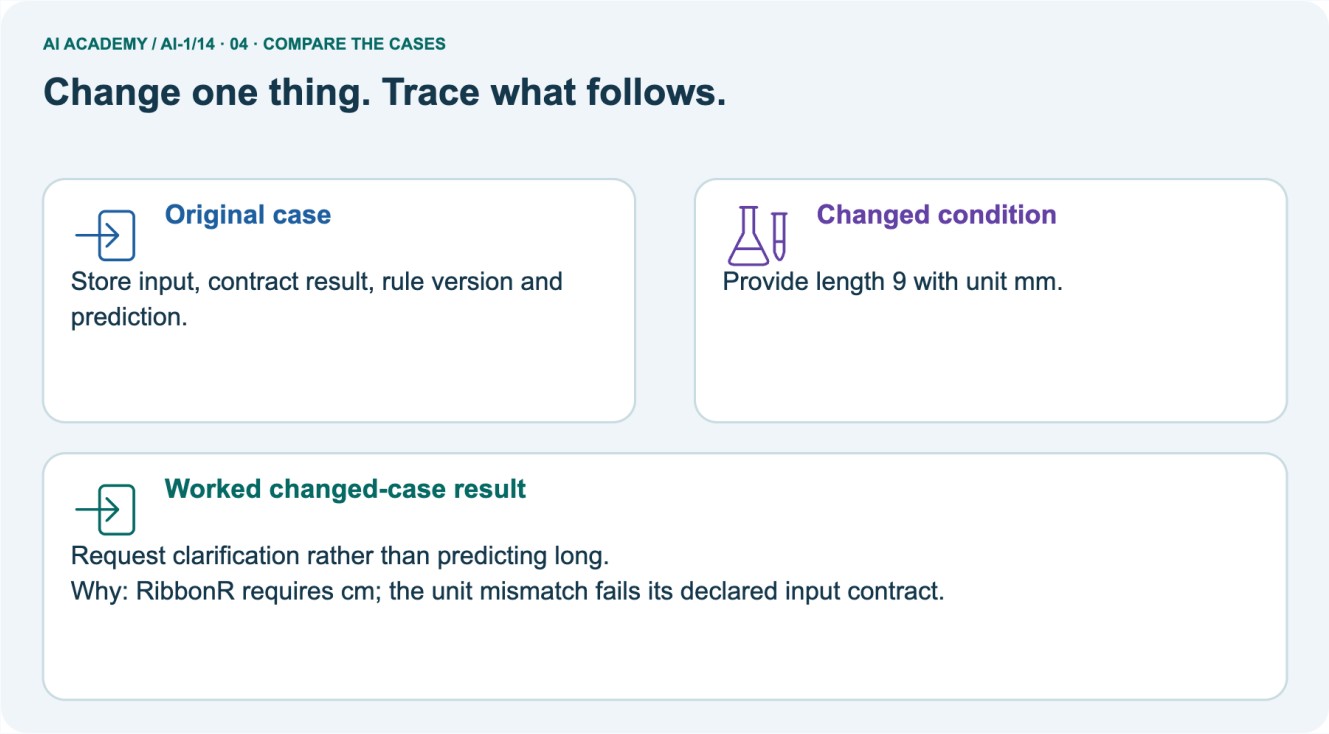
Follow the labelled picture with the matching explanation. Open the vector diagram

Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



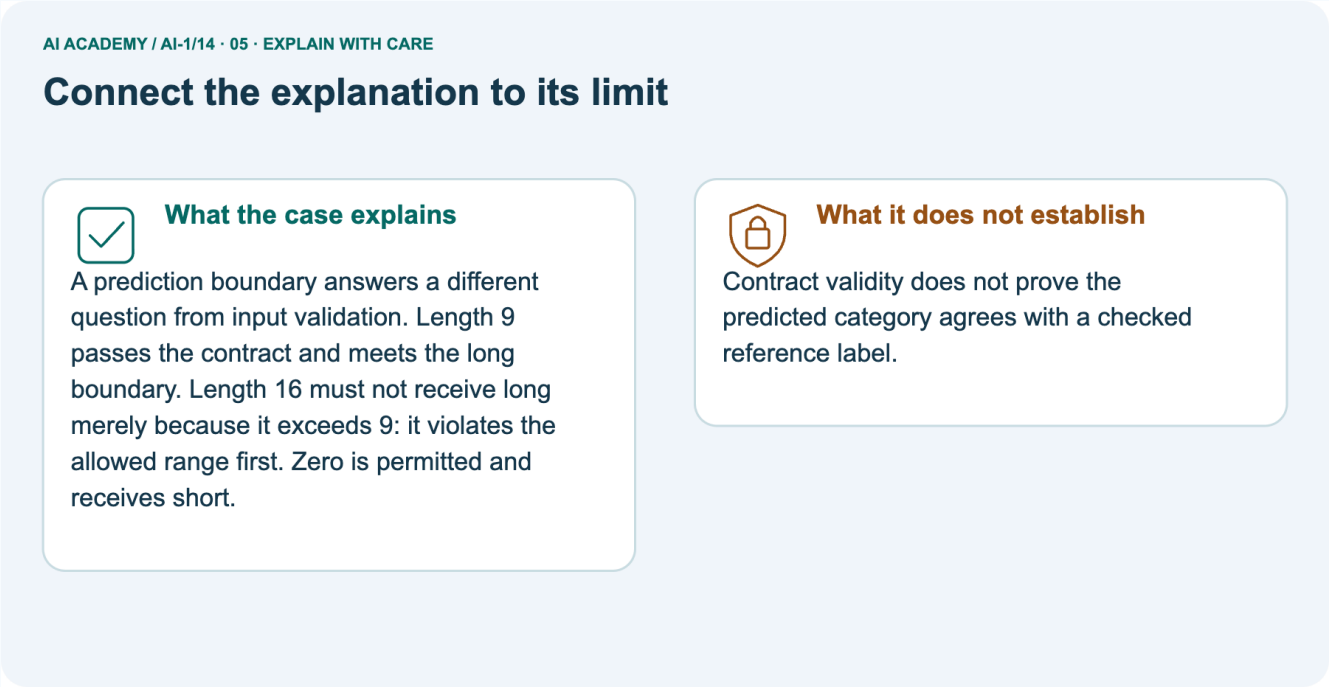
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Store **input**, contract result, rule version and **prediction**. Changed condition: Provide length 9 with unit mm. Expected result: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

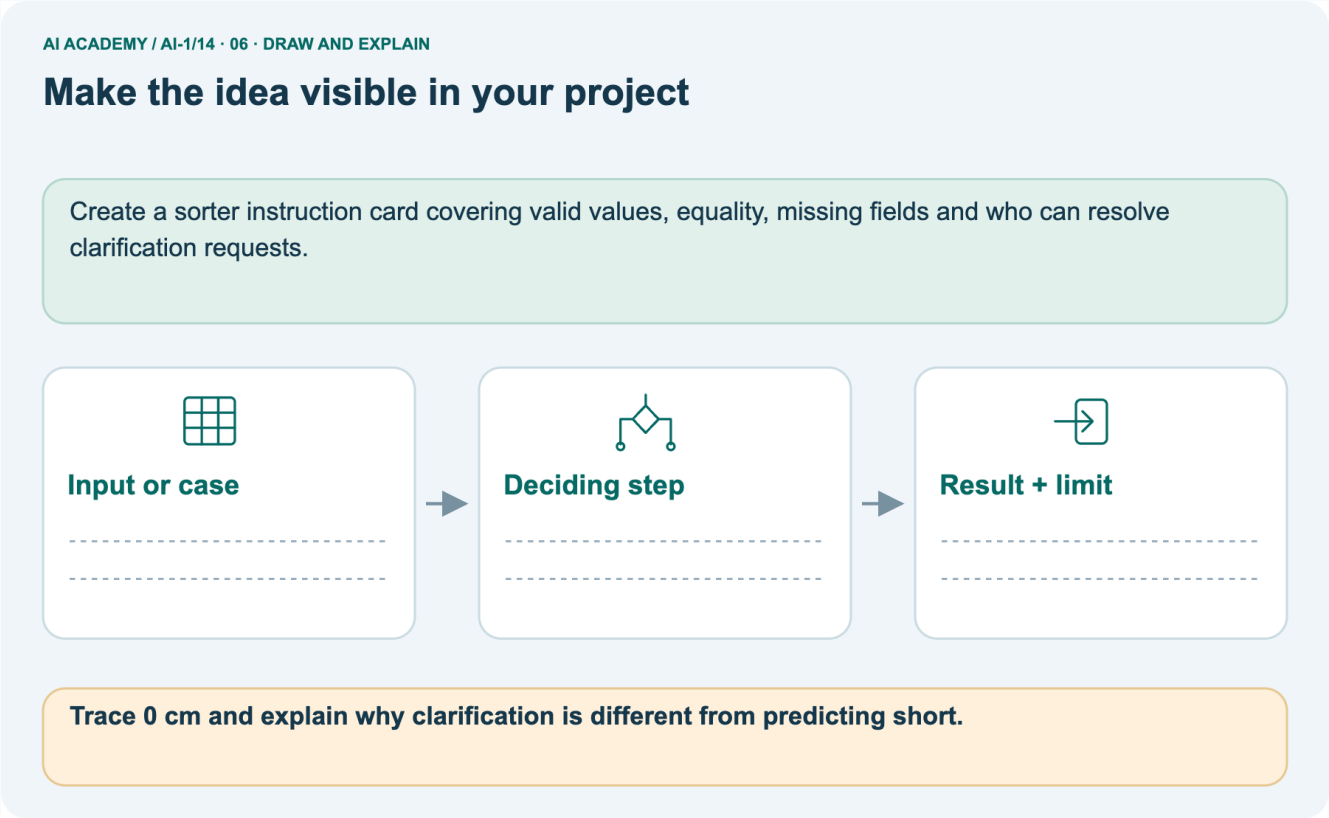
A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Limit: Contract validity does not prove the predicted category agrees with a checked reference label.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests. Draw your **input**, deciding step, result and limit.

Supported question: Trace 0 cm and explain why clarification is different from predicting short.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Build a Human Classifier: the case

02. Follow the deciding steps

03. Three records through the Tower1 classifier

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.